

## **Task 1**

### **Natural Language Processing Skill**

**1. A company is building a Resume Screening System. The system receives raw resume text and**

**must preprocess it to extract meaningful keywords for ranking candidates.**

**You are required to build an NLP pipeline that performs the following steps:**

**Tasks:**

- 1. Tokenize the input text into words**
- 2. Remove stopwords**
- 3. Perform stemming**
- 4. Perform lemmatization**
- 5. Apply POS tagging**
- 6. Compute frequency distribution of remaining words**

### **Coding :**

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer
from nltk import pos_tag, FreqDist

text = input("Enter resume text: ")

tokens = word_tokenize(text)
print("Tokens:", tokens)

stop_words = set(stopwords.words('english'))
words = []

for word in tokens:
    if word.lower() not in stop_words and word.isalpha():
        words.append(word.lower())

print("After Stopword Removal:", words)

ps = PorterStemmer()
stemmed = []

for word in words:
    stemmed.append(ps.stem(word))
```

```
print("Stemmed Words:", stemmed)
```

```
print("POS Tags:", pos_tag(words))
```

```
fd = FreqDist(words)
```

```
print("Frequency Distribution:", fd)
```

Output :

```
Enter resume text:  I am a Python Developer with Machine Learning skills
```

```
Tokens: ['I', 'am', 'a', 'Python', 'Developer', 'with', 'Machine', 'Learning',  
'skills']
```

```
After Stopword Removal: ['python', 'developer', 'machine', 'learning',  
'skills']
```

```
Stemmed Words: ['python', 'develop', 'machin', 'learn', 'skill']
```

```
POS Tags: [('python', 'NN'), ('developer', 'NN'), ('machine', 'NN'),  
(('learning', 'NN'), ('skills', 'NNS'))]
```

```
Frequency Distribution: <FreqDist with 5 samples and 5 outcomes>
```